

Alessandro Sanvito

SOFTWARE AND AI ENGINEER

Amsterdam, Netherlands

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Experience

Optiver

Amsterdam, Netherlands

SOFTWARE ENGINEER

August 2023 - Current

- Expanded the desk beyond index options: built baseline systematic strategies for new product groups from scratch, enabling researchers to rapidly optimize and generate new alpha.
- Selected candidate signals and defined the data-quality and validation criteria each baseline must meet before handoff.
- Built data pipelines processing 10+ TB of instrument-level data with Spark, Polars, and DuckDB, cutting core backtest runtime by 60%; moved ML model training to AWS.

Mercedes-Benz (AI-SEE Project)

Stuttgart, Germany

AI RESEARCH INTERN AND MASTER THESIS STUDENT

May 2022 - June 2023

- Researched NeRFs and diffusion models for 3D scene reconstruction and human avatar modeling from monocular video in autonomous driving systems.
- Outperformed state-of-the-art baselines in PSNR and SSIM when reconstructing scenes in fog, haze, and limited viewpoints; published first-author at IEEE IV 2024 and co-authored at ICCV 2023.

Education

KTH Royal Institute of Technology

Stockholm, Sweden

MSC. IN ICT INNOVATION, DATA SCIENCE

Sept. 2020 - May 2023

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.

Polytechnic University of Milan

Milan, Italy

MSC. IN COMPUTER SCIENCE AND ENGINEERING

Sept. 2020 - Jul. 2023

- 1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on Twitter's 1B-interaction dataset with the university team under prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude

Polytechnic University of Milan

Milan, Italy

BSc. IN ENGINEERING OF COMPUTING SYSTEMS

Sept. 2017 - Jul. 2020

- Final grade: 109/110

Publications

2024	HINT: Learning Complete Human Neural Representations from Limited Viewpoints, Sanvito Alessandro (first author), et al.	IEEE IV 2024
2023	ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings , Ramazzina Andrea, et al., incl. Sanvito Alessandro	ICCV 2023
2022	United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources , Maldini Pietro, Sanvito Alessandro , Surricchio Mattia	ACM recsys'22
2021	Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment , Carminati Luca, et al., incl. Sanvito Alessandro	ACM recsys'21

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, MLLib, AWS
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest